
Higher National Diploma in Data Science 25.1FT
HNDDSFTM1206 - Data Science Research Project
Thesis Report Guidelines(v0.1)

The final report should be approximately 8,000 words and is worth 80% of your total module mark. It must be formatted in Times New Roman, 12-point font, with 1.5 line spacing and 1-inch margins, 16-point font for chapter headings and any other titles 14-point font using IEEE citation style throughout.

1 Front Matter

1. Title Page: Include a concise, descriptive title, student index numbers, supervisor name, and the submission date.
2. Declaration: A signed statement certifying the originality of the work and consent for library use.
3. Abstract (150–250 words): A high-level summary of the research problem, methodology (ex:- Deep Learning, RAG), key results (ex:- WER/CER percentages), and significance.
4. Acknowledgment: Recognition of supervisors, research groups, and others who facilitated the project.
5. Table of Contents: A detailed list of all chapters and sub-sections with page numbers.
6. Lists of Figures, Tables, and Acronyms: Categorized lists for all visual aids and technical abbreviations used in the report

2 Chapter 1: Introduction

1. Background: Provide the broader context of the research topic in data science.
2. Problem Definition: Clearly define the specific challenge or gap the research addresses.
3. Aim, Objectives, and Research Questions: State the broad goal, measurable sub-goals, and 2–4 specific questions your research answers.
4. Key Terms and Concepts: Explain foundational technical concepts (ex:- NLP, ASR, TTS)
5. Scope and Delimitation: Define what the system will do and explicitly state what it will not support.

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6. Significance: Detail the potential impact on inclusivity, accessibility, or the specific field.
 7. Overview of Dissertation: A brief roadmap of the remaining chapters.

3 Chapter 2:Literature Review

1. Domain History: Discuss the evolution of relevant technologies (ex:- the history of ASR or Deep Learning in TTS)
2. Current State-of-the-Art: Review recent studies (ideally post-2020) and use tables to compare techniques and findings.
3. Research Gap: Clearly justify what is missing in existing solutions.

4 Chapter 3:Analysis and Design

1. Research Approach: Define the methodology used, such as Design Science Research (DSR).
2. Data Collection and Preparation: Describe the datasets used, including their size (ex:-hours of speech), variability, and any cleaning or augmentation performed.
3. System Architecture: Provide visual diagrams (Use Case, Activity, and Product Workflow) to show how components interact.
4. Functional and Quality Requirements: List the system's core tasks and performance attributes like usability and scalability.

5 Chapter 4: Implementation

1. Data Processing Pipeline: Detail how inputs are handled (ex:- audio re-sampling to 16kHz)
2. Model Selection and Training: Describe the specific models chosen (ex:- Wav2Vec2-BERT, LLaMA-3.3, VITS) and the transfer learning or fine-tuning process.
3. Integration: Explain how individual modules (ASR, Chatbot, TTS) were unified into a single system.
4. User Interfaces: Showcase the final graphical interfaces for various modules.

6 Chapter 5: Evaluation and Results

1. Quantitative Metrics: Report technical performance using standard metrics like Word Error Rate (WER), Character Error Rate (CER), or Mean Opinion Score (MOS).
2. Qualitative Analysis: Compare model predictions against "Ground Truth" data.
3. Results Comparison: Contrast your results with existing state-of-the-art systems to validate improvements.
4. User Evaluation: Summarize feedback from participants on task success rates and overall satisfaction.

7 Chapter 6: Discussion

1. Research Contributions: Highlight how your work fills the identified gap and its impact on the field.
2. Challenges Faced: Discuss obstacles such as limited computational resources, hardware conflicts, or data scarcity.

8 Chapter 7: Conclusion

1. Summary of Findings: Reiterate how the research questions were answered.
2. Future Work: Suggest specific improvements, such as handling code-mixed speech, improving real-time performance, or reducing model hallucinations.

9 Back Matter

1. References: A minimum of 15–20 recent sources listed in alphabetical order using IEEE style.
2. Appendices: Include supplementary material such as evaluation forms, sample question sets, or code snippets

10 Presentation and Defence

1. Prepare a presentation highlighting the key elements of the project for 15 mins (cut short if exceeded) and followed by a QnA session in front of an examiner.

The End